



GOSFORD HIGH SCHOOL

2009

**HIGHER SCHOOL CERTIFICATE.
HALF YEARLY EXAMINATION.**

MATHEMATICS EXTENSION 2

General Instructions:

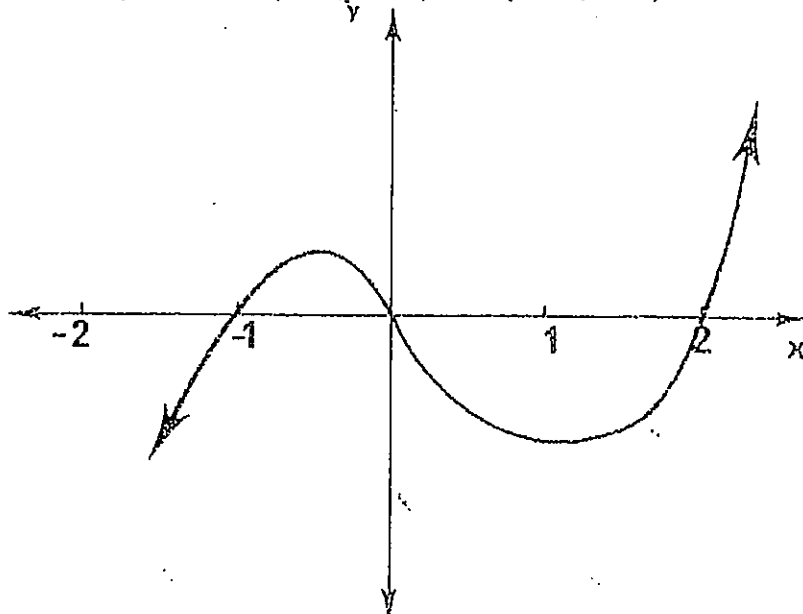
- Reading time – 5minutes.
- Working time – 2hours.
- Write using black or blue pen.
- Board-approved calculators may be used.
- Each question should be started in a new booklet.

Total marks: - 90

- Attempt Questions 1 -4
- All necessary working should be shown.

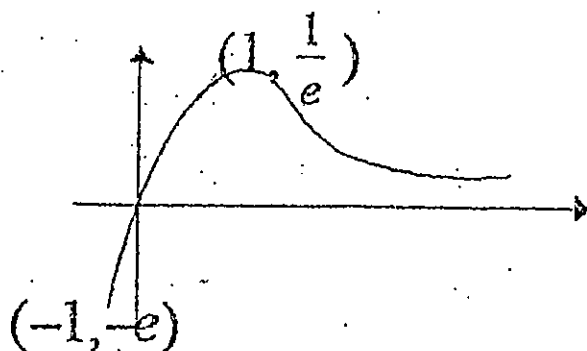
QUESTION 1: (20 marks)

- a. The graph of $f(x) = x^3 - x^2 - 2x$ is given below. Note that the approximate coordinates of the stationary points are $(1.22, -2.11)$ and $(-0.55, 0.63)$



Draw separate one-third page sketches of each of the following:

- i. $y = -f(x)$ (1)
 - ii. $y = |f(x)|$ (1)
 - iii. $y = [f(x)]^2$ (2)
 - iv. $y = \frac{1}{f(x)}$ (2)
 - v. $y = \sqrt{f(x)}$ (2)
- b. i. The graph of $g(x) = xe^{-x}$ for $x \geq -1$ is given below:



Draw separate sketches showing $g(x-2)$ and $g(-x)$ (2)

(Question 1 continued)

- ii. The function $f(x)$ is given by

$$f(x) = g(x-2), \quad x \geq 1$$

$$= g(-x), \quad x < 1$$

Draw a neat sketch of $y = f(x)$ showing all important features. (2)

- c. i. Show that if $f(x) = \frac{2x^2-1}{x^2-1}, x \neq \pm 1$, then $f(x)$ is an even function. (1)

- ii. Sketch the curve $y = \frac{2x^2-1}{x^2-1}, x \neq \pm 1$, showing the location and nature of all stationary points, the equations of all asymptotes and any intercepts with the co-ordinate axes. (7)

Question 2: (Start a new booklet) (20 marks)

- a. i. For the complex number $z = 1 - \sqrt{3}i$ find:

$$|z| \text{ and } \arg(z) \quad (1)$$

- ii. Express $\bar{z}$, z^2 and $\frac{1}{z}$ in the form $a+ib$ where a, b are real numbers (3)

- iii. Plot $z, \bar{z}, z^2$ and $\frac{1}{z}$ on an Argand Diagram (1)

- b. Solve $z^2 + 4z - 1 + 12i = 0$ (3)

- c. Sketch on separate Argand Diagrams each of the following regions:

i. $2 < |z| \leq 3$ and $\frac{\pi}{4} \leq \arg(z) \leq \frac{3\pi}{4}$ (2)

ii. $|z - 2i| > |z - 2 + i|$ (2)

iii. $z\bar{z} \leq z + \bar{z}$ (2)

- d. Use De Moivre's Theorem to express $\cos 5\theta$ and $\sin 5\theta$ in powers of $\sin \theta$ and $\cos \theta$. Hence express $\tan 5\theta$ as a rational function of t where $t = \tan \theta$. Deduce that:

$$\tan \frac{\pi}{5} \tan \frac{2\pi}{5} \tan \frac{3\pi}{5} \tan \frac{4\pi}{5} = 5 \quad (6)$$

Question 3: (Start a new booklet) (25 marks)

- a. Without evaluating the integral explain why

$$\int_{-1}^1 x e^{-x^2} dx = 0 \quad (1)$$

b. i. Find $\int \frac{\cos^3 x}{\sin^2 x} dx$ (3)

ii. Evaluate $\int_{-1}^2 x \sqrt{2-x} dx$ (3)

- c. Use the substitution $u = \sin 2x$ or otherwise to evaluate

$$\int_0^{\pi/4} \frac{\sin 4x}{1 + \sin^2 2x} dx \quad (3)$$

d. Show that $\int_0^{\pi/4} x \sec^2 x dx = \frac{1}{2} \left(\frac{\pi}{2} - \ln 2 \right)$ (4)

- e. i. Use the method of partial fractions to find real numbers A, B and C such that

$$\frac{x}{(x-1)^2(x-2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x-2} \quad (2)$$

- ii. Hence show that

$$\int_0^{1/2} \left(\frac{x}{(x-1)^2(x-2)} \right) dx = 2 \log_e \left(\frac{3}{2} \right) - 1 \quad (3)$$

f. i. Given that $I_{2n+1} = \int_0^1 x^{2n+1} e^{x^2} dx$

show that $I_{2n+1} = \frac{1}{2} e - n I_{2n-1}$ (3)

- ii. Hence or otherwise evaluate

$$\int_0^1 x^5 e^{x^2} dx \quad (3)$$

Question 4: (Start a new booklet) (25 marks)

- a. i. Show that $x - 3$ is a factor of the polynomial

$$Q(x) = 4x^3 - 15x^2 + 8x + 3 \quad (1)$$

- ii. Given that the equation $x^4 - 5x^3 + 4x^2 + 3x + 9 = 0$ has a root of multiplicity 2, solve the equation completely. (6)

- b. Show that $1+i$ is a zero of the polynomial $P(x) = x^3 - x^2 + 2$ and hence resolve $P(x)$ into irreducible factors over the field of

- i. real numbers
ii. complex numbers (5)

- c. When a polynomial $P(x)$ is divided by $(x - 2)$ and $(x - 3)$ the respective remainders are 4 and 9. Find the remainder when $P(x)$ is divided by $(x - 2)(x - 3)$ (4)

- d. The equation $2x^3 - 3x - 1 = 0$ has three non zero roots α, β and γ . Evaluate

i. $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ (1)

ii. $\alpha^4 + \beta^4 + \gamma^4$ (4)

- e. If one root of the equation $x^3 - px^2 + qx - r = 0$ is equal to the product of the other two roots prove that:

$$r(p+1)^2 = (q+r)^2 \quad (4)$$

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

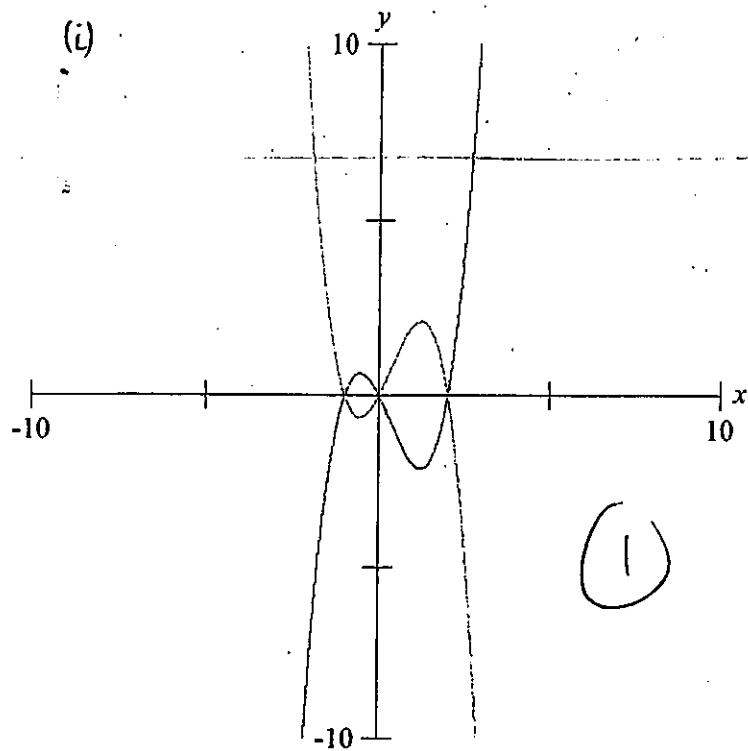
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

1. α)

17x2 H/Y

(i)

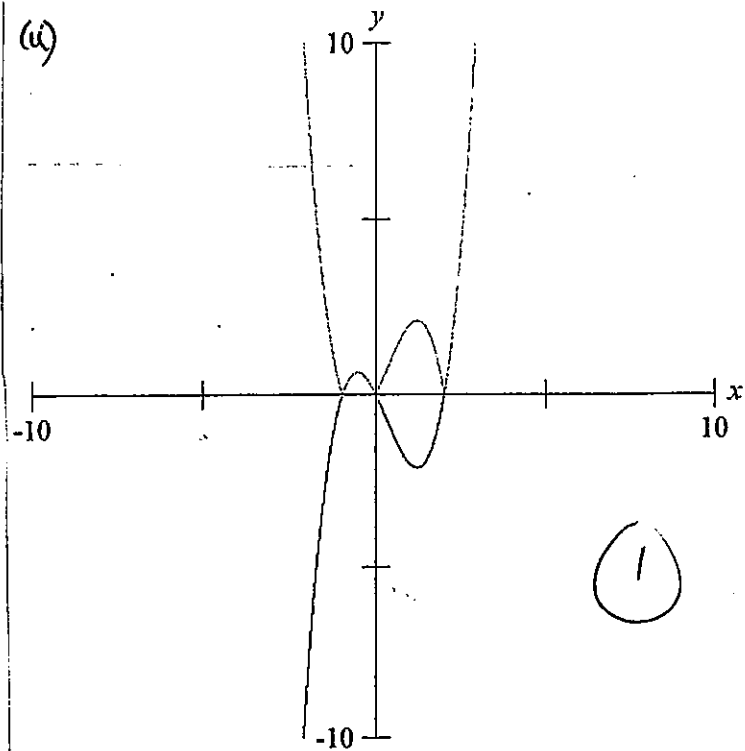


(1)

$$y = x^3 - x^2 - 2x$$

$$y = -(x^3 - x^2 - 2x)$$

(ii)

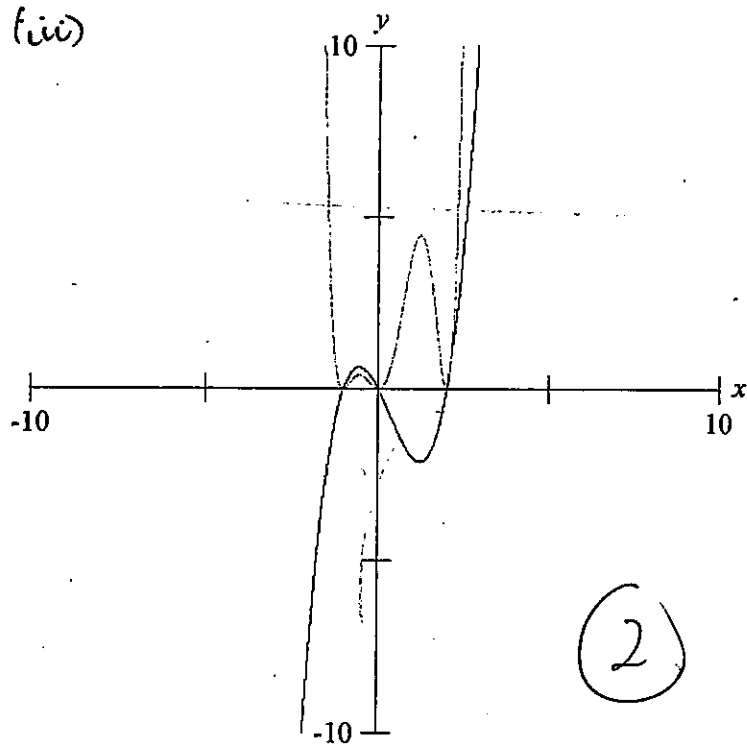


(1)

$$y = x^3 - x^2 - 2x$$

$$y = |x^3 - x^2 - 2x|$$

(iii)

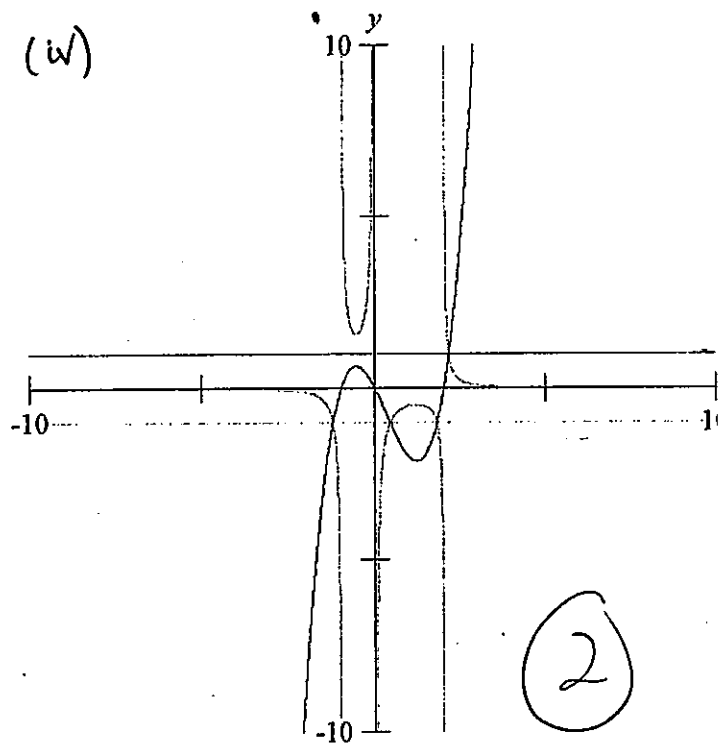


(2)

$$y = x^3 - x^2 - 2x$$

$$y = (x^3 - x^2 - 2x)^2$$

(iv)



(2)

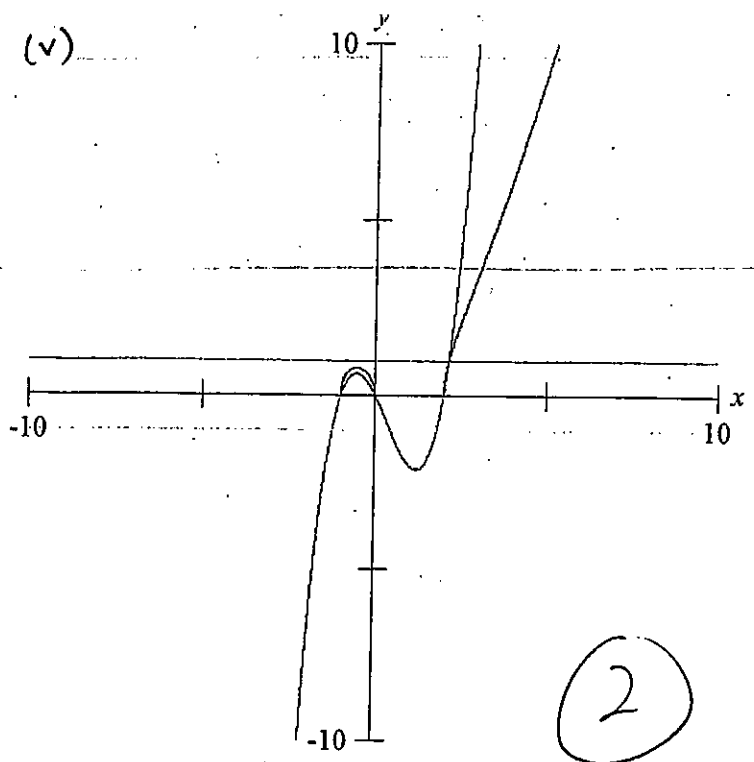
$$y = x^3 - x^2 - 2x$$

$$y = \frac{1}{(x^3 - x^2 - 2x)}$$

$$y = 1$$

$$y = -1$$

(v)



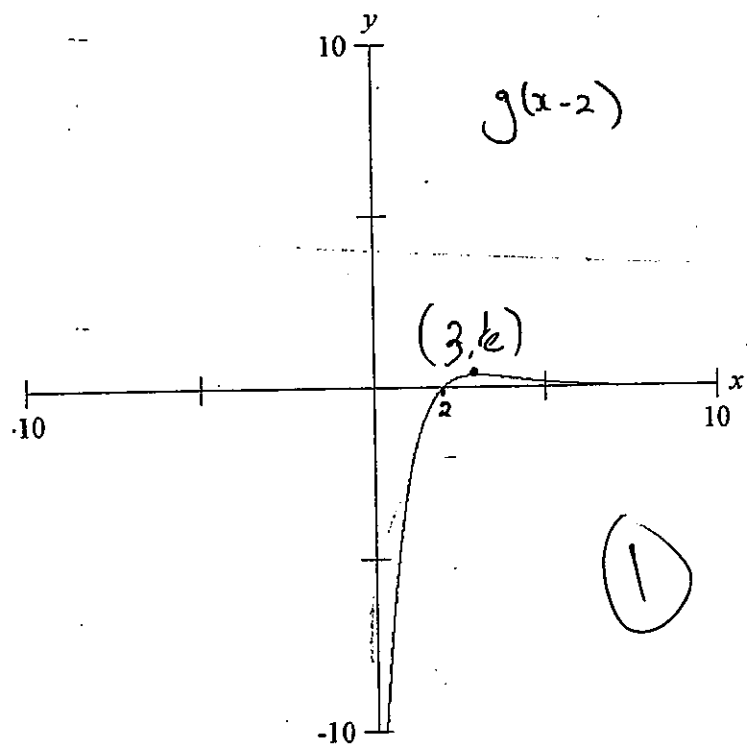
$$y = x^3 - x^2 - 2x$$

$$y = \sqrt{(x^3 - x^2 - 2x)}$$

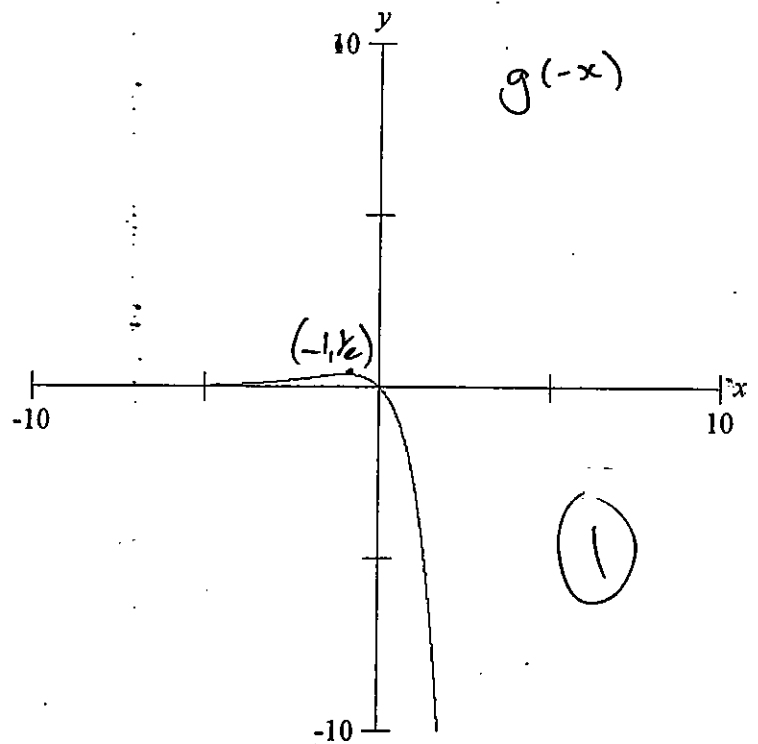
$$y = 1$$

$$y = -1$$

b) c)

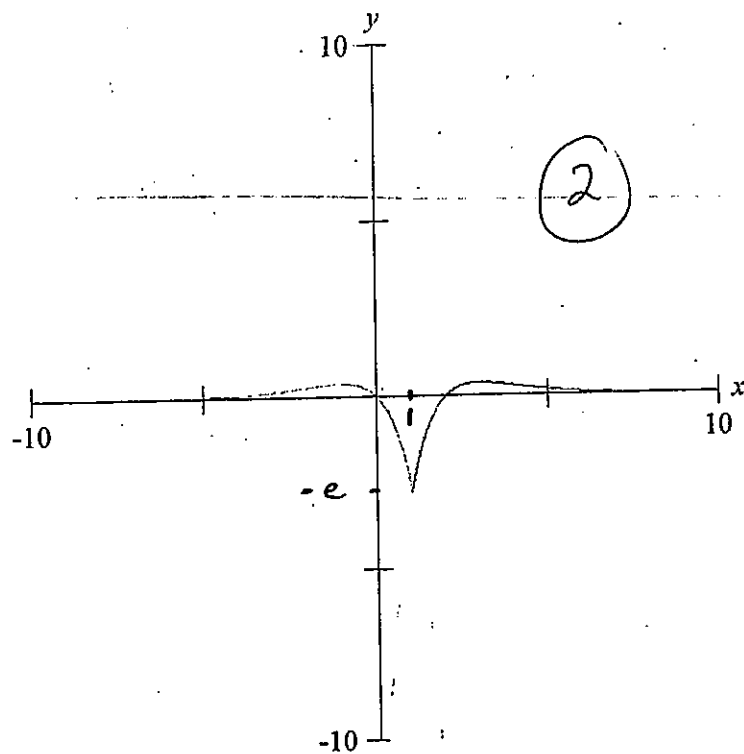


$$y = (x-2) \cdot e^{-(x-2)}$$



$$y = -x \cdot e^x$$

(c)



$$y = (x-2)e^{-(x-2)}$$

$$y = -xe^x$$

c) (i) $f(x) = \frac{2x^2 - 1}{x^2 - 1}, x \neq \pm 1$

$$\begin{aligned} f(-x) &= \frac{2(-x)^2 - 1}{(-x)^2 - 1} \\ &= \frac{2x^2 - 1}{x^2 - 1} \end{aligned}$$

(1)

$$\therefore f(x) = f(-x)$$

$\therefore$ hence $f(x)$ is an even function.

ii) If $y = \frac{2x^2 - 1}{x^2 - 1}$

$$= \frac{2x^2 - 2}{x^2 - 1} + \frac{1}{x^2 - 1}$$

$$= 2 + \frac{1}{x^2 - 1}$$

(2)

$\therefore$ Horizontal asymptote at $y = 2$.

Vertical asymptotes at $x = \pm 1$

$$I \int y = 2 + (x^2 - 1)^{-1}$$

$$y' = -1(x^2 - 1)^{-2} \cdot 2x$$

$$= \frac{-2x}{(x^2 - 1)^2}$$

When $y' = 0$, $-2x = 0$
 $x = 0$
 $\Delta y = 1.$

x	$0 - \varepsilon$	0	$0 + \varepsilon$
y'	$+$	0	$-$

(2)

$(0, 1)$ is a max t.p.

Also when $y = 0$, $\frac{2x^2 - 1}{x^2 - 1} = 0$

$$\therefore x^2 = \frac{1}{2}$$

$$x = \pm \frac{1}{\sqrt{2}}$$

(1)

As $x \rightarrow 1^+$, $y \rightarrow \infty$

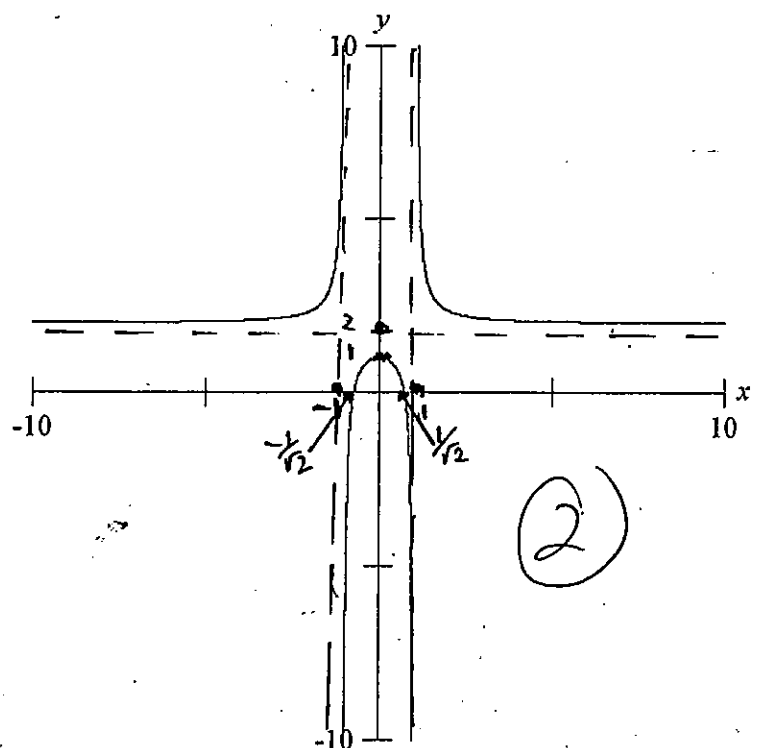
As $x \rightarrow 1^-$, $y \rightarrow -\infty$

As $x \rightarrow -1^+$, $y \rightarrow -\infty$

As $x \rightarrow -1^-$, $y \rightarrow \infty$

As $x \rightarrow \infty$, $y \rightarrow 2^+$

As $x \rightarrow -\infty$, $y \rightarrow 2^+$



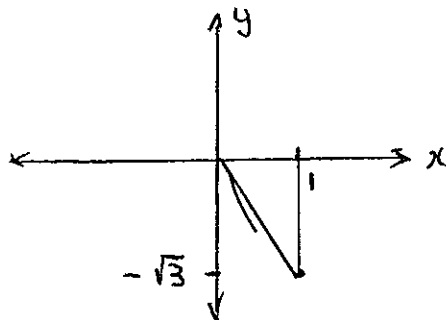
2) a)

$$(i) |z| = \sqrt{1^2 + (-\sqrt{3})^2}$$

$$= 2$$

(1)

$$\text{Arg}(z) = -\frac{\pi}{3}$$



$$(ii) \bar{z} = 1 + \sqrt{3}i \quad (1)$$

$$z^2 = (1 - \sqrt{3}i)^2$$

$$= 1 - 2\sqrt{3}i + 3i^2$$

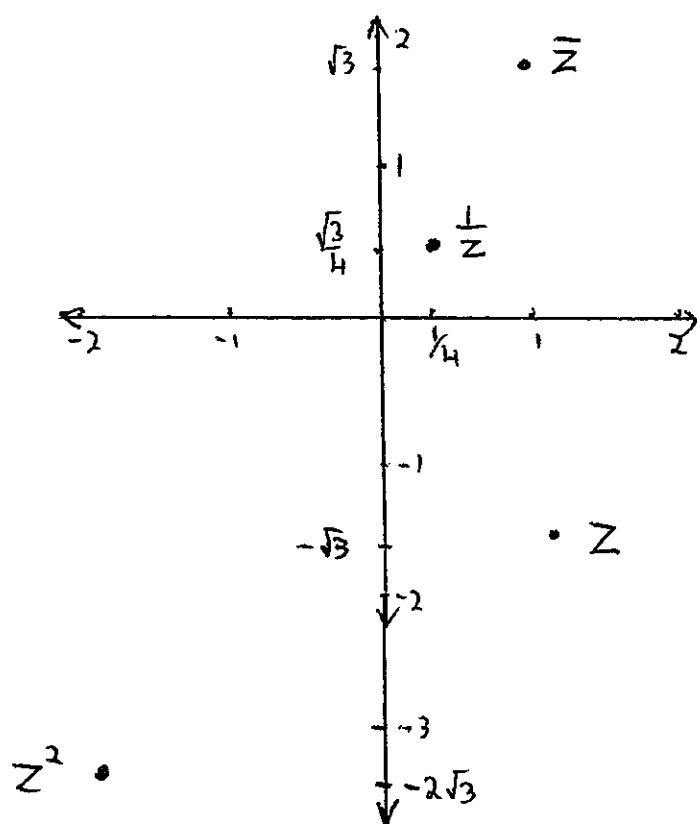
$$= -2 - 2\sqrt{3}i \quad (1)$$

$$\frac{1}{z} = \frac{1}{1 - \sqrt{3}i} \times \frac{1 + \sqrt{3}i}{1 + \sqrt{3}i}$$

$$= \frac{1 + \sqrt{3}i}{1 - (\sqrt{3}i)^2}$$

$$= \frac{1 + \sqrt{3}i}{4} \quad (1)$$

(iii)



(1)

$$b) \quad z^2 + 4z - 1 + 12i = 0$$

$$z = \frac{-4 \pm \sqrt{16 - 4(-1 + 12i)}}{2}$$

$$= \frac{-4 \pm \sqrt{20 - 48i}}{2}$$

$$= -2 \pm \sqrt{5 - 12i}$$

$$\text{Let } \sqrt{5 - 12i} = a + ib$$

$$a^2 - b^2 = 5 \quad \& \quad 2ab = -12$$

$$ab = -6$$

(3)

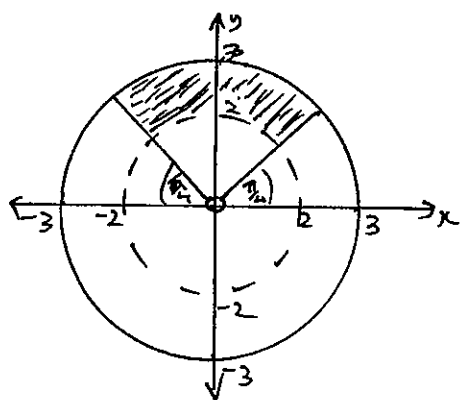
By inspection $a = \pm 3$, $b = \mp 2$.

$$\therefore \sqrt{5 - 12i} = \pm (3 - 2i)$$

$$\therefore z = -2 \pm (3 - 2i)$$

$$= 1 - 2i \quad \text{or} \quad -5 + 2i$$

c) (i)



(2)

$$(ii) \quad |z - 2i| > |z - 2 + i|$$

$$\text{Let } z = x + iy$$

$$\begin{aligned} |x + iy - 2i| &= |x + i(y - 2)| \\ &= \sqrt{x^2 + (y - 2)^2} \end{aligned}$$

$$|x+iy-2+i| = |(x-2)+i(y+1)|$$

$$= \sqrt{(x-2)^2 + (y+1)^2}$$

$$\therefore x^2 + (y-2)^2 > (x-2)^2 + (y+1)^2$$

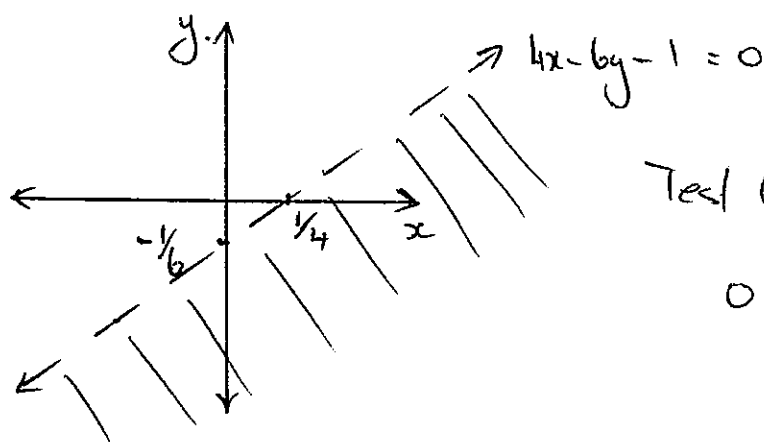
$$x^2 + y^2 - 4y + 4 > x^2 - 4x + 4 + y^2 + 2y + 1$$

$$\therefore 4x - 6y - 1 > 0$$

$$\text{If } 4x - 6y - 1 = 0$$

$$6y = 4x - 1$$

$$y = \frac{2}{3}x - \frac{1}{6}$$



(2)

Test (0,0)

$$0 - 0 - 1 \neq 0$$

$$(iii) \quad z\bar{z} \leq z + \bar{z}$$

$$\text{Let } z = x+iy$$

$$(x+iy)(x-iy) \leq x+iy + x-iy$$

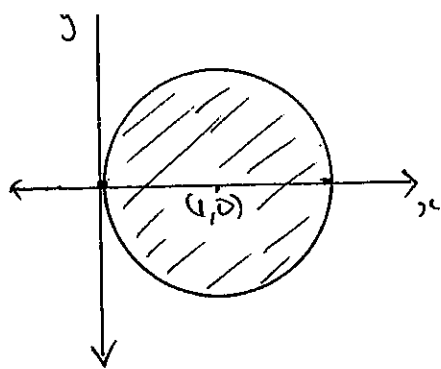
$$x^2 + y^2 \leq 2x$$

$$x^2 - 2x + y^2 \leq 0$$

$$x^2 - 2x + 1 + y^2 \leq 1$$

$$(x-1)^2 + y^2 \leq 1$$

Circle centre (1,0) & radius 1.



(2)

$$d) (\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$$

$$\text{LHS} = \cos^5 \theta + 5 \cos^4 \theta i \sin \theta + 10 \cos^3 \theta i^2 \sin^2 \theta + 10 \cos^2 \theta i^3 \sin^3 \theta + 5 \cos \theta i^4 \sin^4 \theta + i^5 \sin^5 \theta$$

$$= \cos^5 \theta + 5 \cos^4 \theta \sin \theta i - 10 \cos^3 \theta \sin^2 \theta - 10 \cos^2 \theta \sin^3 \theta i + 5 \cos \theta \sin^4 \theta + \sin^5 \theta i$$

Equating real and imaginary parts

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

$$\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$$

$$\therefore \tan 5\theta = \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$$

Divide throughout by $\cos^5 \theta$

$$\tan 5\theta = \frac{5 \frac{\sin \theta}{\cos \theta} - 10 \frac{\sin^3 \theta}{\cos^3 \theta} + \frac{\sin^5 \theta}{\cos^5 \theta}}{1 - 10 \frac{\sin^2 \theta}{\cos^2 \theta} + 5 \frac{\sin^4 \theta}{\cos^4 \theta}}$$

$$= \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4} \quad \text{where } t = \tan \theta$$

If $\tan 5\theta = 0$

$$\frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4} = 0$$

$$\therefore 5t - 10t^3 + t^5 = 0$$

$$\text{i.e. } t^5 - 10t^3 + 5t = 0$$

Now if $\tan 5\theta = 0$, the "fired" 5 solns are

$$\tan 0, \tan \frac{\pi}{5}, \tan \frac{2\pi}{5}, \tan \frac{3\pi}{5} \text{ \& } \tan \frac{4\pi}{5}$$

$$\therefore t = 0, \tan \frac{\pi}{5}, \tan \frac{2\pi}{5}, \tan \frac{3\pi}{5}, \tan \frac{4\pi}{5}.$$

Also if $t^5 - 10t^3 + 5t = 0$

$$t(t^4 - 10t^2 + 5) = 0$$

$$\therefore t = 0 \text{ \& the roots of } t^4 - 10t^2 + 5 = 0$$

$$\text{are } \tan \frac{\pi}{5}, \tan \frac{2\pi}{5}, \tan \frac{3\pi}{5}, \tan \frac{4\pi}{5}$$

Now the product of the roots is $\frac{e}{a}$

$$\therefore \tan \frac{\pi}{5} \cdot \tan \frac{2\pi}{5} \cdot \tan \frac{3\pi}{5} \cdot \tan \frac{4\pi}{5} = \frac{5}{1}$$

$$\text{i.e. } \tan \frac{\pi}{5} \cdot \tan \frac{2\pi}{5} \cdot \tan \frac{3\pi}{5} \cdot \tan \frac{4\pi}{5} = 5.$$

6

3/ a) Let $f(x) = x e^{-x^2}$

$$f(-x) = -x e^{-(x)^2}$$

$$= -x e^{-x^2}$$

$$= -f(x)$$

$\therefore f(x)$ is odd.

(1)

Hence $\int_{-1}^1 x e^{-x^2} dx = 0.$

b) (i) Let $I = \int \frac{\cos^2 x}{\sin^2 x} dx$

Let $u = \sin x$

$$du = \cos x dx$$

$$I = \int \frac{\cos^2 x}{\sin^2 x} \cdot \cos x dx$$

$$= \int \frac{1 - \sin^2 x}{\sin^2 x} \cdot \cos x dx$$

$$= \int \frac{1 - u^2}{u^2} du$$

$$= \int u^{-2} - 1 du$$

$$= \frac{u^{-1}}{-1} - u + C$$

$$= \frac{-1}{\sin x} - \sin x + C.$$

(3)

ii) Let $I = \int_{-1}^2 x \sqrt{2-x} dx$

Let $u^2 = 2 - x$

$$2u du = -dx$$

If $x = -1, u = \sqrt{3}$

$x = 2, u = 0.$

$$I = \int_{\sqrt{3}}^0 (2-u^2) \cdot u \cdot -2u \, du$$

$$= -2 \int_{\sqrt{3}}^0 2u^2 - u^4 \, du$$

$$= -2 \left[\frac{2u^3}{3} - \frac{u^5}{5} \right]_{\sqrt{3}}^0$$

$$= -2 \left\{ 0 - \left(2 \times \frac{3\sqrt{3}}{3} - \frac{9\sqrt{3}}{5} \right) \right\}$$

3

$$= -2 \left(-2\sqrt{3} + \frac{9\sqrt{3}}{5} \right)$$

$$= -2 \left(\frac{-10\sqrt{3} + 9\sqrt{3}}{5} \right)$$

$$= \frac{2\sqrt{3}}{5}$$

c) Let $I = \int_0^{\pi/4} \frac{\sin 4x}{1 + \sin^2 2x} \, dx$

Let $u = \sin 2x$

$du = 2 \cos 2x \, dx$

If $x = 0$, $u = 0$

$x = \frac{\pi}{4}$, $u = 1$.

$$I = \int_0^{\pi/4} \frac{2 \sin 2x \cos 2x}{1 + \sin^2 2x} \, dx$$

$$= \int_0^1 \frac{u}{1+u^2} \, du$$

3

$$= \frac{1}{2} \left[\ln(1+u^2) \right]_0^1$$

$$= \frac{1}{2} \ln 2 - \frac{1}{2} \ln 1$$

$$= \frac{1}{2} \ln 2 \quad \text{OR} \quad \ln \sqrt{2}$$

$$d) \text{ Let } I = \int_0^{\pi/4} x \sec^2 x \, dx$$

$$u = x, \quad v' = \sec^2 x$$

$$u' = 1, \quad v = \tan x$$

$$I = \left[x \tan x \right]_0^{\pi/4} - \int_0^{\pi/4} 1 \cdot \tan x \, dx$$

$$= \left(\frac{\pi}{4} \cdot 1 - 0 \right) - \int_0^{\pi/4} \frac{\sin x}{\cos x} \, dx$$

$$= \frac{\pi}{4} + \int_0^{\pi/4} \frac{-\sin x}{\cos x} \, dx$$

$$= \frac{\pi}{4} + \left[\ln(\cos x) \right]_0^{\pi/4}$$

$$= \frac{\pi}{4} + \left(\ln \frac{1}{\sqrt{2}} - \ln 1 \right)$$

$$= \frac{\pi}{4} + \ln(2)^{-1/2} - 0$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2$$

$$= \frac{1}{2} \left(\frac{\pi}{2} - \ln 2 \right)$$

(4)

$$e) \text{ (i) } \int \frac{x}{(x-1)^2(x-2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x-2}$$

$$x = A(x-1)(x-2) + B(x-2) + C(x-1)^2$$

$$\text{Let } x=1, \quad 1 = 0 - B + 0$$

$$\therefore B = -1$$

$$\text{Let } x=2, \quad 2 = 0 + 0 + C$$

$$C = 2$$

$$\text{Let } x=0, \quad 0 = 2A - 2B + C$$

$$0 = 2A + 2 + 2$$

$$A = -2$$

(2)

$$(ii) \text{ Let } I = \int_0^{\frac{1}{2}} \frac{x}{(x-1)^2(x-2)} dx$$

$$= \int_0^{\frac{1}{2}} \frac{-2}{x-1} - \frac{1}{(x-1)^2} + \frac{2}{x-2} dx$$

$$= \int_0^{\frac{1}{2}} \frac{-2}{x-1} - (x-1)^{-2} + \frac{2}{x-2} dx$$

$$= \left[-2 \ln(x-1) - \frac{(x-1)^{-1}}{-1 \times 1} + 2 \ln(x-2) \right]_0^{\frac{1}{2}}$$

$$= \left[2 \ln \left[\frac{x-2}{x-1} \right] + \frac{1}{x-1} \right]_0^{\frac{1}{2}}$$

$$= \left(2 \ln \left(\frac{-\frac{1}{2}}{-\frac{1}{2}} \right) + \frac{1}{-\frac{1}{2}} \right) - \left(2 \ln \left(\frac{-2}{-1} \right) + \frac{1}{-1} \right)$$

$$= 2 \ln 3 - 2 - 2 \ln 2 + 1$$

$$= 2 \ln \left(\frac{3}{2} \right) - 1$$

(3)

$$f) I_{2n+1} = \int_0^1 x^{2n+1} e^{x^2} dx$$

$$u = x^{2n}, \quad v' = x e^{x^2}$$

$$u' = 2n x^{2n-1}, \quad v = \frac{1}{2} e^{x^2}$$

$$I_{2n+1} = \left[\frac{1}{2} x^{2n} e^{x^2} \right]_0^1 - \int_0^1 \frac{1}{2} e^{x^2} \cdot 2n x^{2n-1} dx$$

$$= \left(\frac{1}{2} \cdot 1 \cdot e - \frac{1}{2} \cdot 0 \cdot 1 \right) - \int_0^1 n x^{2n-1} e^{x^2} dx$$

$$= \frac{e}{2} - n I_{2n-1}$$

(3)

$$ii) \text{ N.B : If } 2n+1 = 5, \quad n=2.$$

$$\therefore I_5 = \frac{e}{2} - 2 I_3.$$

$$\text{If } 2n+1 = 3, \quad n=1$$

$$I_3 = \frac{e}{2} - 2 \left[\frac{e}{2} - 1 \cdot I_1 \right]$$

$$\text{If } 2n+1 = 1, \quad n=0$$

$$\therefore I_1 = \frac{e}{2} - e + 2 \int_0^1 x e^{x^2} dx$$

$$= -\frac{e}{2} + 2 \left[\frac{1}{2} e^{x^2} \right]_0^1$$

$$= -\frac{e}{2} + (e - 1)$$

$$= \frac{e}{2} - 1.$$

(3)

4. a)

$$\begin{aligned} \text{(i)} \quad Q(3) &= 4 \times (3)^3 - 15 \times (3)^2 + 8 \times (3) + 3 \\ &= 4 \times 27 - 15 \times 9 + 8 \times 3 + 3 \\ &= 0. \end{aligned}$$

(1)

$\therefore x-3$ is a factor of $Q(x)$.

$$\text{(ii)} \quad \text{Let } P(x) = x^4 - 5x^3 + 4x^2 + 3x + 9$$

$$P'(x) = 4x^3 - 15x^2 + 8x + 3.$$

$$\text{From (i)} \quad P'(3) = 0$$

$$\begin{aligned} \text{Also } P(3) &= (3)^4 - 5 \times (3)^3 + 4 \times (3)^2 + 3 \times (3) + 9. \\ &= 81 - 5 \times 27 + 4 \times 9 + 3 \times 3 + 9 \\ &= 0. \end{aligned}$$

$\therefore x-3$ is a double root of $P(x)$

$$\begin{array}{r} x^2 + x + 1 \\ x^2 - 6x + 9 \overline{) x^4 - 5x^3 + 4x^2 + 3x + 9} \\ \underline{x^4 - 6x^3 + 9x^2} \\ x^3 - 5x^2 + 3x \\ \underline{x^3 - 6x^2 + 9x} \\ x^2 - 6x + 9 \\ \underline{x^2 - 6x + 9} \\ 0 \end{array}$$

(6)

$$\therefore P(x) = (x-3)^2 (x^2 + x + 1)$$

$$\text{If } (x-3)^2 (x^2 + x + 1) = 0$$

$$(x-3)^2 \left(x^2 + x + \frac{1}{4} + \frac{3}{4} \right) = 0$$

$$(x-3)^2 \left[\left(x + \frac{1}{2} \right)^2 - \frac{3}{4} i^2 \right] = 0$$

$$(x-3)^2 \left[\left(x + \frac{1}{2} + \frac{\sqrt{3}}{2} i \right) \left(x + \frac{1}{2} - \frac{\sqrt{3}}{2} i \right) \right] = 0$$

$$\therefore x = 3, -\frac{1}{2} - \frac{\sqrt{3}}{2} i, -\frac{1}{2} + \frac{\sqrt{3}}{2} i$$

$$b) \quad P(1+i) = (1+i)^3 - (1+i)^2 + 2$$

$$= 1 + 3i + 3i^2 + i^3 - (1 + 2i + i^2) + 2$$

$$= 1 + 3i - 3 - i - 1 - 2i + 1 + 2$$

$$= 0$$

$\therefore (1+i)$ is a zero of $P(x)$

Now, if $(1+i)$ is a zero so is $(1-i)$

$$\begin{aligned} \therefore [x - (1+i)][x - (1-i)] &= x^2 - (1-i)x - (1+i)x + (1+i)(1-i) \\ &= x^2 - x + ix - x - ix + 1 - i^2 \\ &= x^2 - 2x + 2 \end{aligned}$$

$$\begin{array}{r} x + 1. \\ x^2 - 2x + 2 \overline{) x^3 - x^2 + 0x + 2} \\ \underline{x^3 - 2x^2 + 2x} \\ x^2 - 2x + 2 \end{array}$$

(5)

$$(i) \quad \therefore P(x) = (x^2 - 2x + 2)(x + 1)$$

$$\begin{aligned} (ii) \quad P(x) &= [x - (1+i)][x - (1-i)](x + 1) \\ &= (x - 1 - i)(x - 1 + i)(x + 1) \end{aligned}$$

$$c) \quad (x-2)(x-3) = x^2 - 5x + 6$$

$$\therefore P(x) = (x^2 - 5x + 6) \cdot Q(x) + R(x)$$

where the degree of $R(x)$ is less than the degree of $x^2 - 5x + 6$.

$\therefore R(x)$ is of the form $ax + b$.

$$\text{Now } P(2) = 4.$$

$$\therefore 4 = 0 \cdot Q(x) + 2a + b$$

$$\text{i.e. } 2a + b = 4 \quad \text{--- (i)}$$

$$\therefore P(3) = 9.$$

$$\therefore 9 = 0 \cdot Q(x) + 3a + b$$

$$3a + b = 9 \quad \text{--- (2)}$$

(4)

$$\textcircled{2} - \textcircled{1}$$

$$a = 5$$

$$\therefore b = -6$$

Hence the remainder is $5x - 6$

$$d) \quad 2x^3 - 3x - 1 = 0$$

$$\therefore 2x^3 + 0x^2 - 3x - 1 = 0$$

$$(i) \quad \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= \frac{\frac{c}{a}}{-\frac{b}{a}}$$

$$= \frac{-\frac{3}{2}}{\frac{1}{2}}$$

$$= -3$$

(1)

$$(iii) \quad \text{If } 2x^3 - 3x - 1 = 0$$

$$2x^3 = 3x + 1.$$

If α, β, γ are zeros of $P(x)$

$$2\alpha^3 = 3\alpha + 1$$

$$2\beta^3 = 3\beta + 1$$

$$2\gamma^3 = 3\gamma + 1$$

Multiply by α, β & γ respectively

$$2\alpha^4 = 3\alpha^2 + \alpha \quad \text{--- (1)}$$

$$2\beta^4 = 3\beta^2 + \beta \quad \text{--- (2)}$$

$$2\gamma^4 = 3\gamma^2 + \gamma \quad \text{--- (3)}$$

$$(1) + (2) + (3)$$

$$\begin{aligned} 2(\alpha^4 + \beta^4 + \gamma^4) &= 3(\alpha^2 + \beta^2 + \gamma^2) + \alpha + \beta + \gamma \\ &= 3[(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)] + \alpha + \beta + \gamma \\ &= 3\left[\left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right)\right] + \left(-\frac{b}{a}\right) \\ &= 3\left[0^2 - 2 \times -\frac{3}{2}\right] + 0 \\ &= 9. \end{aligned}$$

(4)

$$\therefore \alpha^4 + \beta^4 + \gamma^4 = \frac{9}{2}.$$

e) Let the roots be α, β & $\alpha\beta$

$$\text{Now } \sum \alpha = -\frac{b}{a}$$

$$\therefore \alpha + \beta + \alpha\beta = p \quad \text{--- (1)}$$

$$\sum \alpha\beta = \frac{c}{a}$$

$$\alpha\beta + \alpha^2\beta + \alpha\beta^2 = q \quad \text{--- (2)}$$

$$\sum \alpha\beta\gamma = -\frac{d}{a}$$

$$\alpha^2\beta^2 = r \quad \text{--- (3)}$$

From ①

$$p + 1 = 1 + \alpha + \beta + \alpha\beta$$

$$\therefore r(p+1)^2 = \alpha^2\beta^2(1+\alpha+\beta+\alpha\beta)$$

From ② & ③

$$\begin{aligned}(q+r)^2 &= (\alpha\beta + \alpha^2\beta + \alpha\beta^2 + \alpha^2\beta^2)^2 \\&= [\alpha\beta(1+\alpha+\beta+\alpha\beta)]^2 \\&= \alpha^2\beta^2(1+\alpha+\beta+\alpha\beta)^2 \\&= r(p+1)^2.\end{aligned}$$

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